ChatGPT

We first note that the given code defines a computable real number a_2. Analyzing the series shows

$$f(\infty) = 72^{-10^{24}} + \sum_{k=3}^{\infty} \left((\tfrac{1}{2})^k - 72^{-10^{24}k} \right) \approx \tfrac{1}{4} + O(10^{-10^{24}})\,,$$

so $a_2 = 2f(\infty) + \frac{1}{2} = 1 + 2 \cdot 72^{-10^{24}} - o(72^{-10^{24}})$. In other words, a_2 is essentially 1 (slightly above 1). (Indeed, “computable numbers” like this are defined by such algorithms [1], but here the tiny correction is negligible.)

Thus $a_2 > a_1 = 1$, and we have the linear constraint $x_1 + a_2 x_2 = 1$. To minimize $x_1 + x_2$ under $x_1 \geq 0, x_2 \geq 0$, one uses only the faster therapy (rate a_2). Hence the optimum is $x_1 = 0$, $x_2 = 1/a_2$. Numerically $1/a_2 \approx 1/(1+\epsilon) \approx 1$ (since $\epsilon = 2 \cdot 72^{-10^{24}}$ is astronomically small). To two-decimal precision in the sup norm, the minimiser is

$$x_1 \approx 0.00, \qquad x_2 \approx 1.00.$$

This yields total time $0.00 + 1.00 = 1.00$, which is minimal.

Answer: $x_1 \approx 0.00,\ x_2 \approx 1.00$ (to two-decimal accuracy).

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[1] Computable number - Wikipedia
https://en.wikipedia.org/wiki/Computable_number